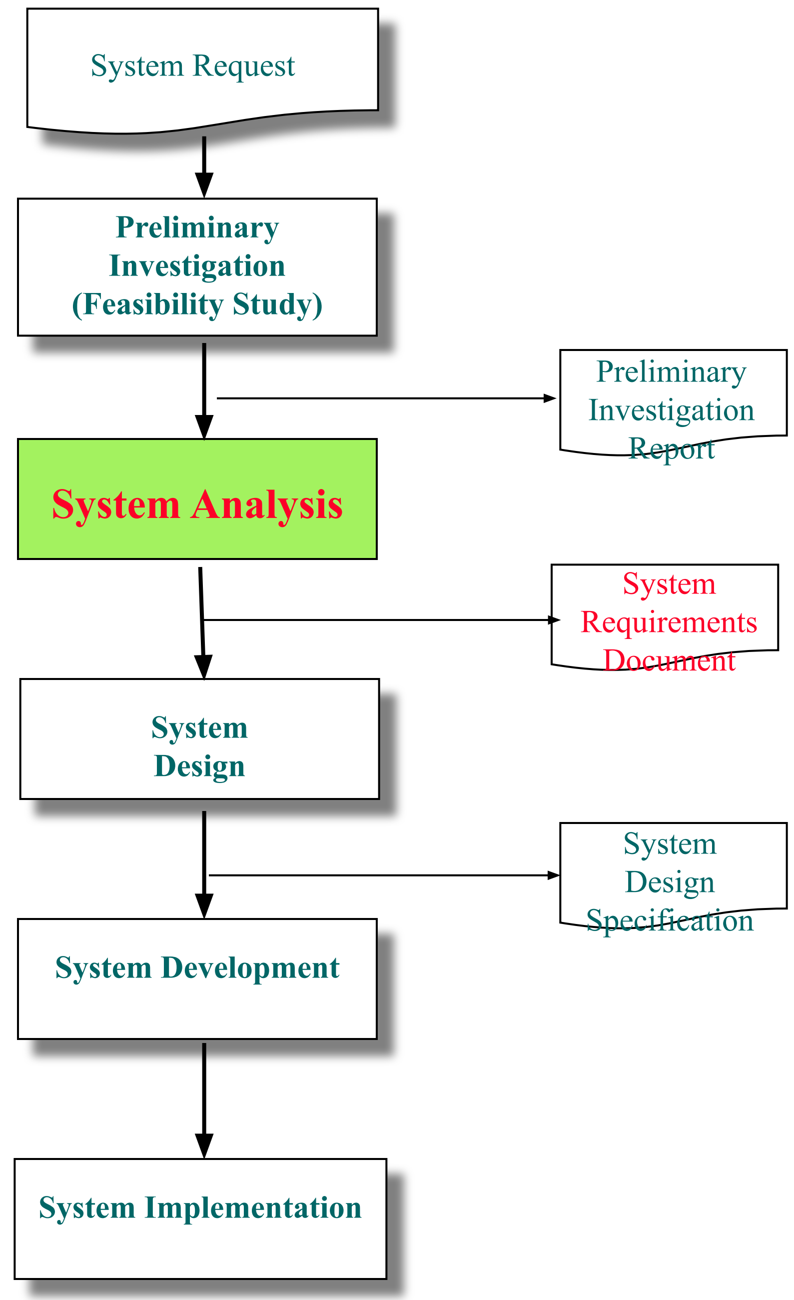
Chapter 5

System Analysis 2

- Fact Recording 2

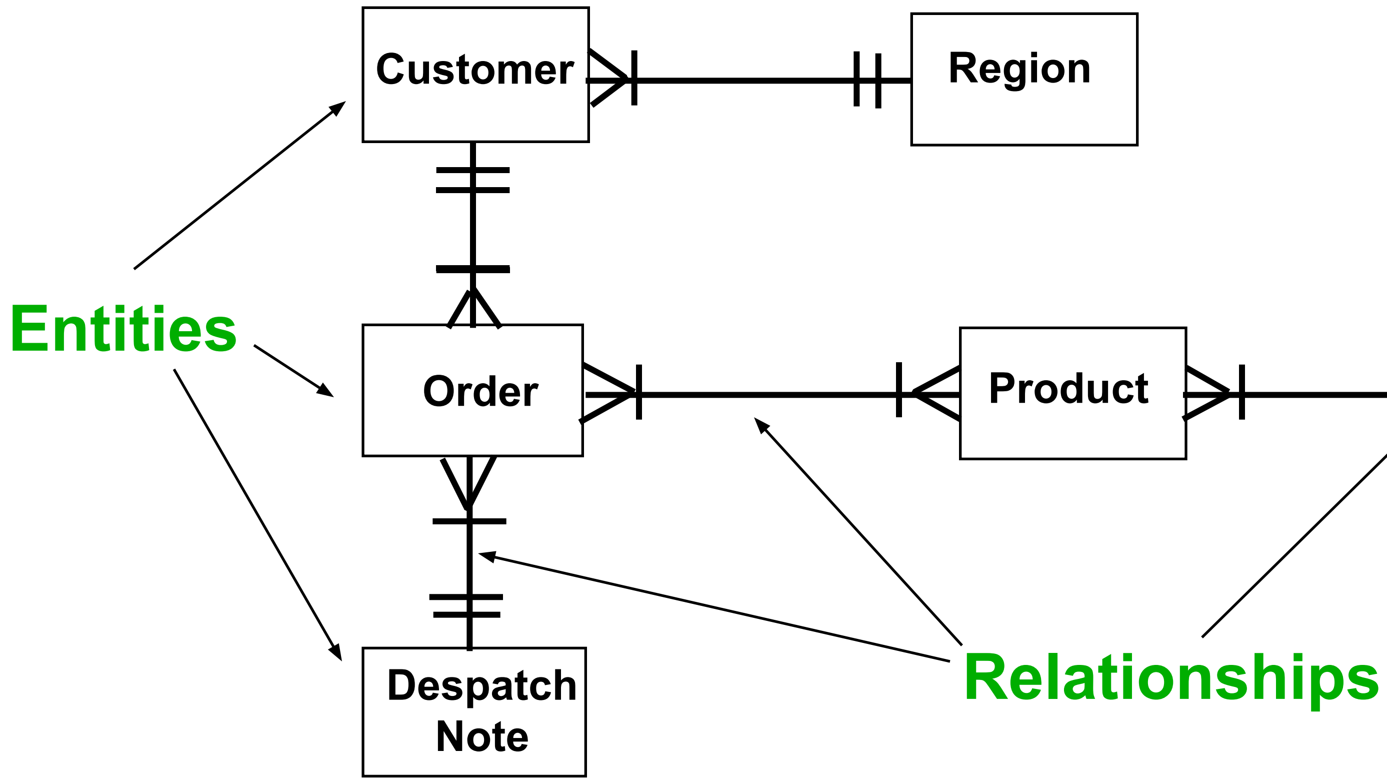
SDLC


System Request
Preliminary Investigation (Feasibility Study)
Preliminary Investigation Report
System Analysis
System Requirements Document
System Design
System Design Specification
System Development
System Implementation

Entity Relationship Diagram (ERD)

Components of ERD

- There are only 2 components :



ERD Notations

	Crow's Foot	Chen
Entity		
Relationship line		
Relationship	None	
Option Symbol		
One (1) symbol		1
One and only one symbol		1
Many (M) symbol		M (or N)

Note : This University adopts the *Crow's Foot* notation for all its *tests*, *assignments* and examination, for this course.

Entity (= file, table, data stored)

- **Meaning**. A class of persons, places, branches, departments, objects, or transactions. They store data which are entered or captured. Described using a noun.

- **Examples** : customers, products, suppliers, PO, payment, SO, quotations.

- **Notation**. It is represented by a rectangular box.

Examples of Entities Types

- **Persons** : contractor, customer, employee, student, supplier, publisher, applicant.
- **Objects** : book, machine, stock (part, product, item), course, programme.
- **Events/Transactions** : application, reservation, cancellation, class, flight, invoice, registration, payment, renewal, manpower requisition, reservation, SO, PO, RFQ, quotation.
- **Places** : sales region, building, room, branch office, campus, department, faculty

Exercise 2

- List 4 appropriate entities for each of the following systems :
  - Inventory System
  - Recruitment System
  - Purchasing System

Answers 2

- Inventory System
  - Supplier, Item, Goods Delivered, Goods Received etc
- Recruitment System
  - Department, Manpower Requisition, Application, Applicant, Interview, Job etc
- Purchasing System
  - Supplier, Purchaser, Item, Purchase Order, Quotation, RFQ etc

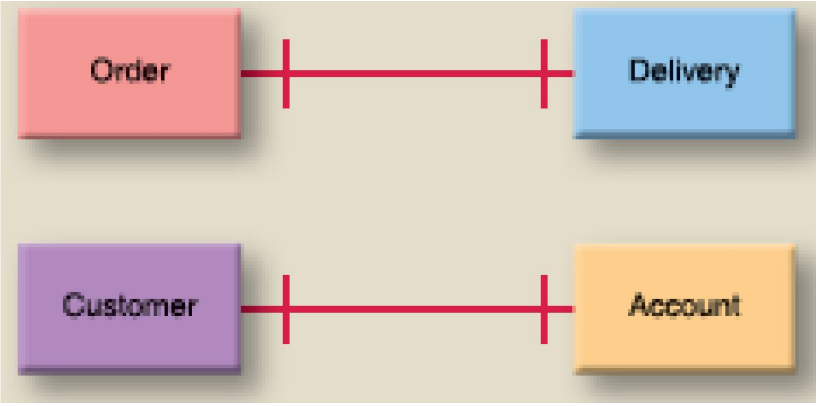
Note : the answers above are NOT exhaustive, just samples only

Relationship

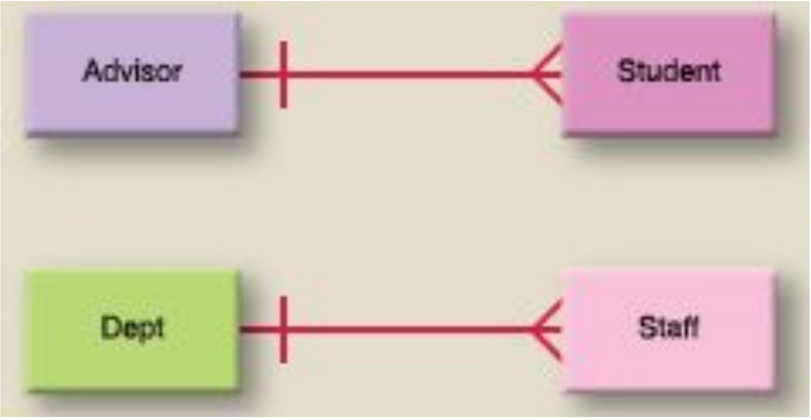
- **Meaning.** **The lines between the entities (boxes) represent relationships.**
- **Example.** **Between SUPPLIER and PURCHASE ORDER**
- **Notation.** **The relationship is indicated by lines connecting the entities**

3 Types of Relationship

- One-to-One (1 : 1)

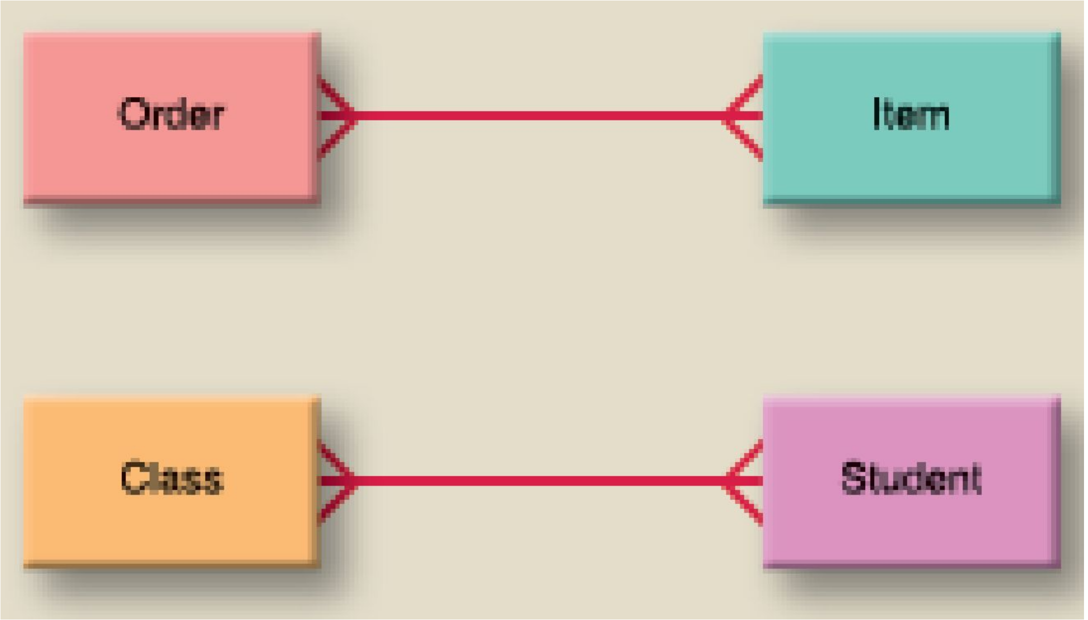


- One-to-Many (1 : N)



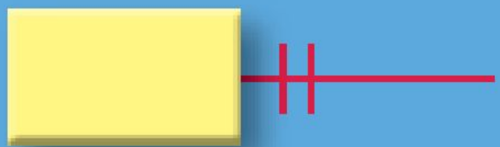
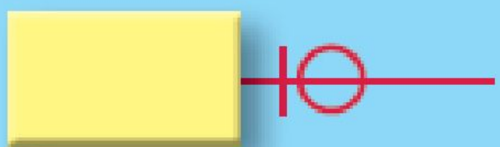
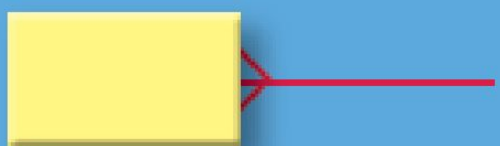
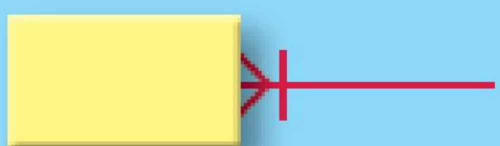
Types of Relationship

- **Many-to-Many (M : N)**



Cardinality – no. of occurrences

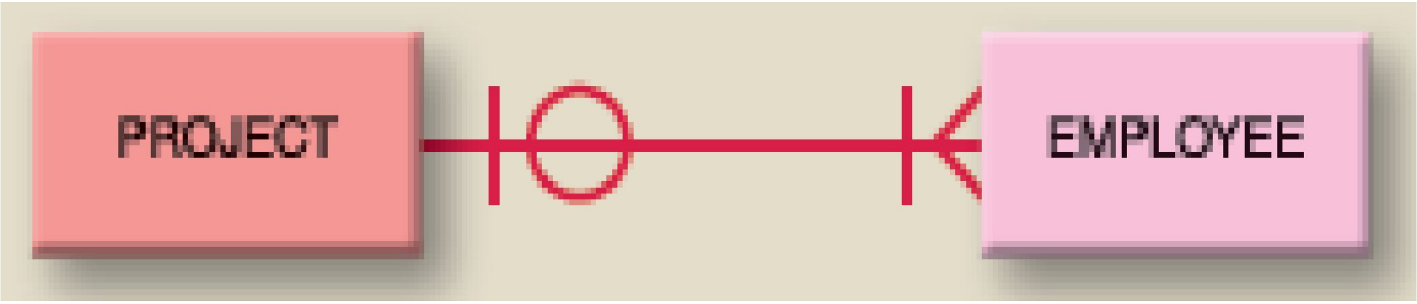
Crow's Foot Notation

Symbols	Meaning
	One and only one
	Zero or one
	More than one
	one or more than one
	Zero or one or more than one (ie. More than zero)

Note : This University adopts the Crow's Foot notation for all its tests, assignments and examination, for this course.

Crow's Foot Notation

- Describes the type of relationship between two entities
- Cardinality must be defined in both directions for every relationship.



1st direction → One Project has *One or Many* Employee

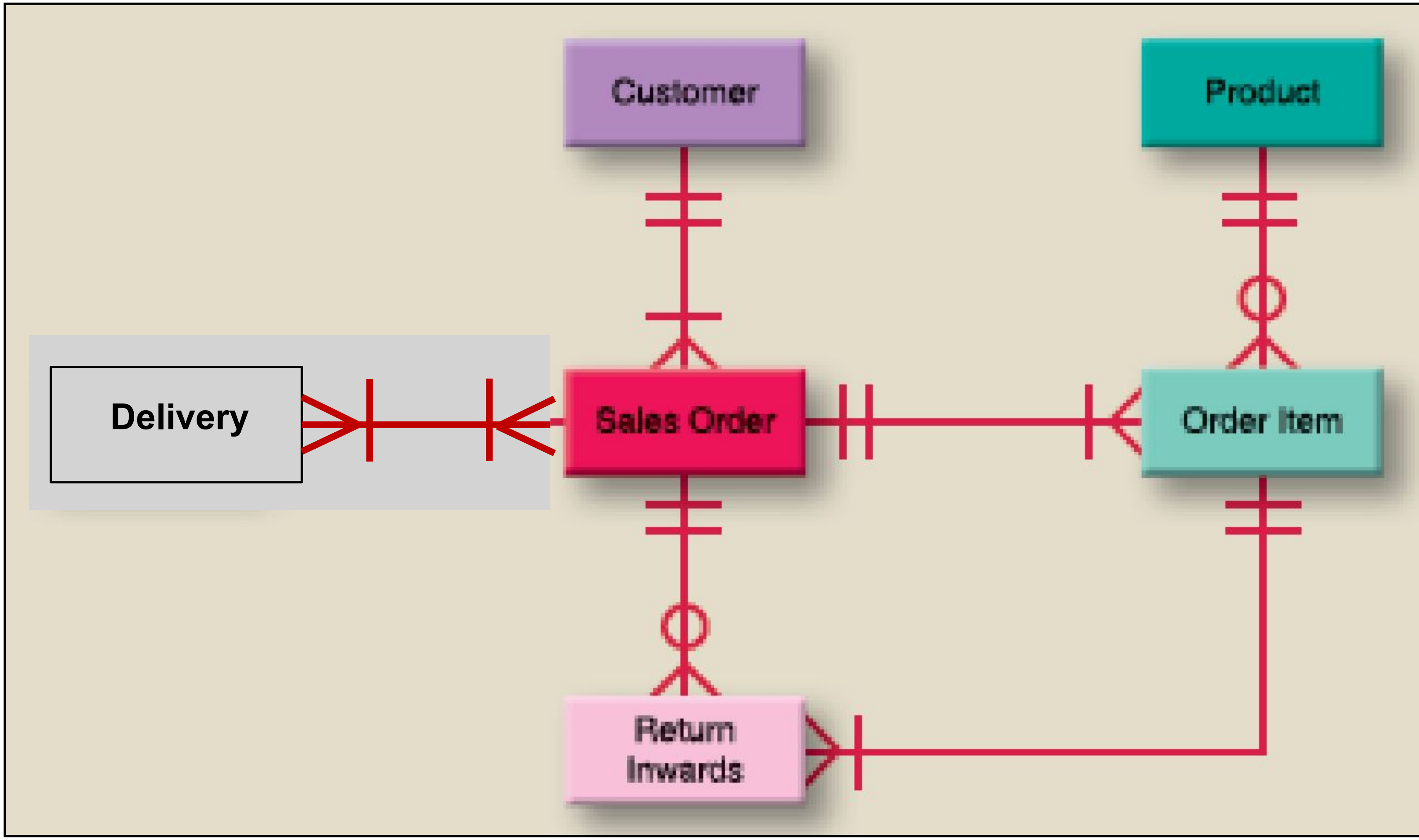
2nd direction → One Employee takes part in *Zero or One* Project

Crow's Foot Notation Examples

One ~~and only one~~ CUSTOMER can open zero or one or many ACCOUNT.
One ACCOUNT is opened by only one CUSTOMER

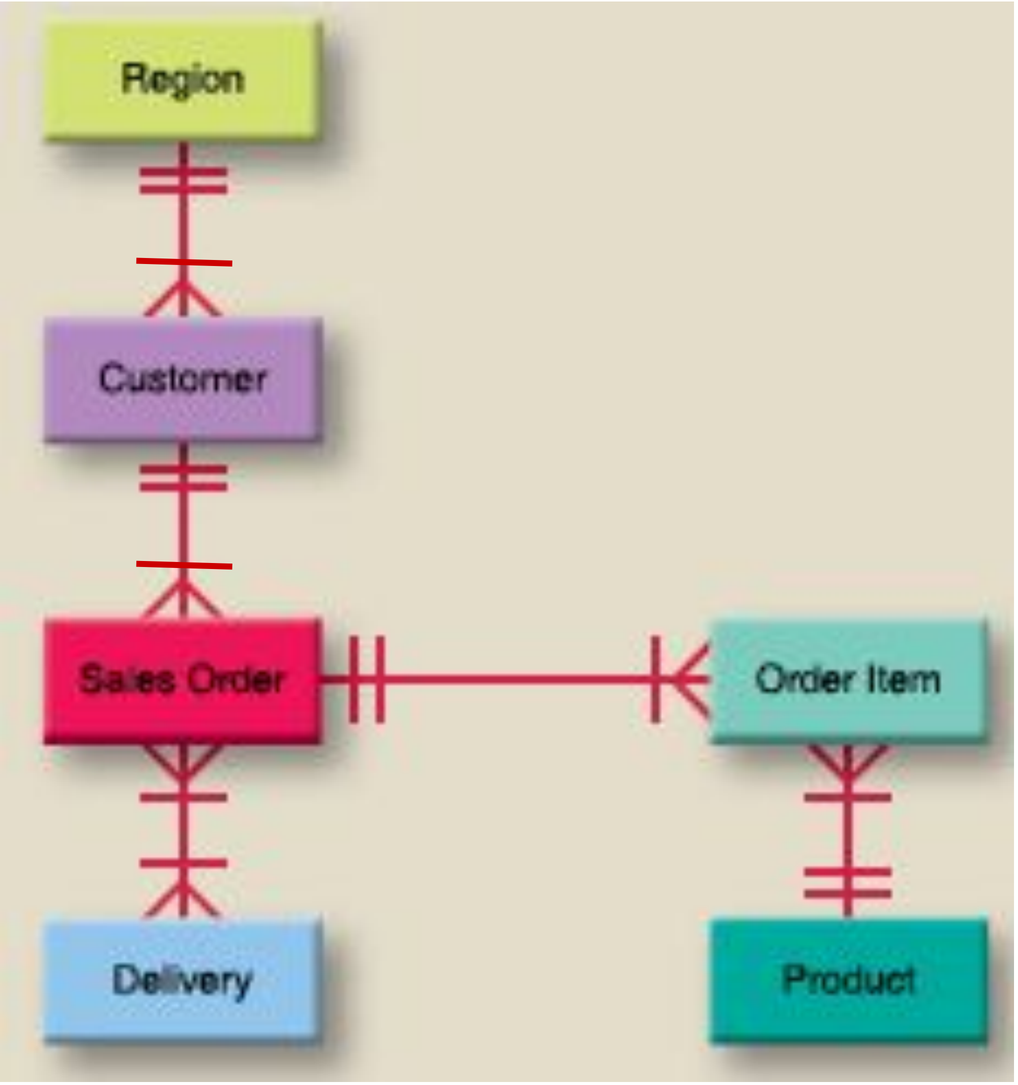
One STUDENT can apply for zero or one LOAN.
One LOAN is applied by zero or one, or many STUDENT

Example 1 - Inventory System

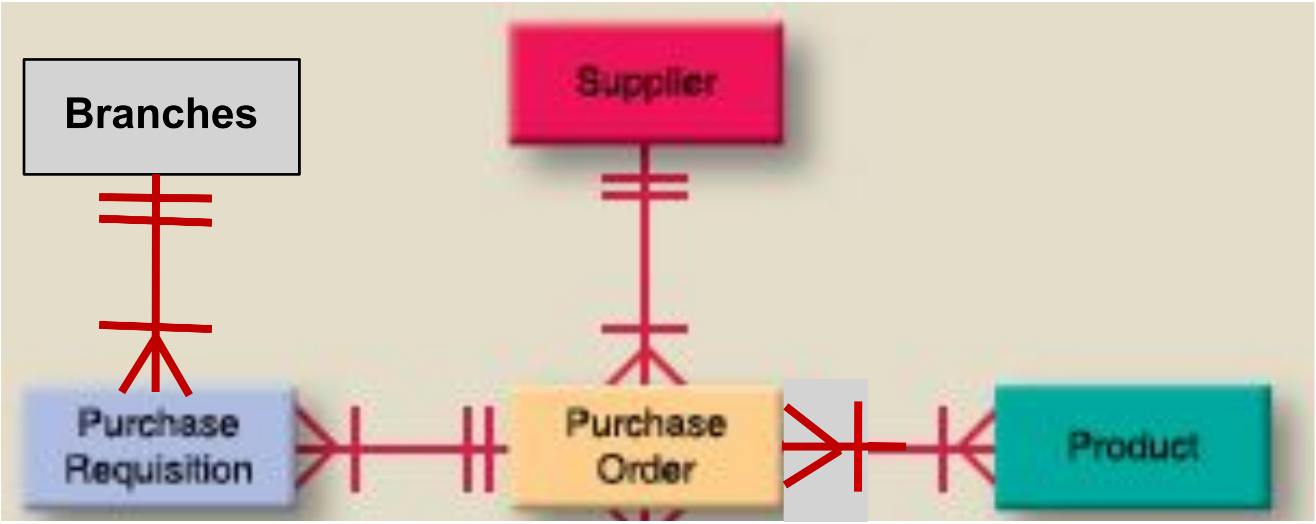


Try reading each of the above 6 pairs of relationships

Example 2 – Sales System



Example 3 – Purchase Order Processing



Try to read each of the above 4 pairs of relationships
Try to identify ONE error from the above relationships

Resolving
Many-to-Many Relationship

(**Note** : any many-to-many relationships must be resolved)

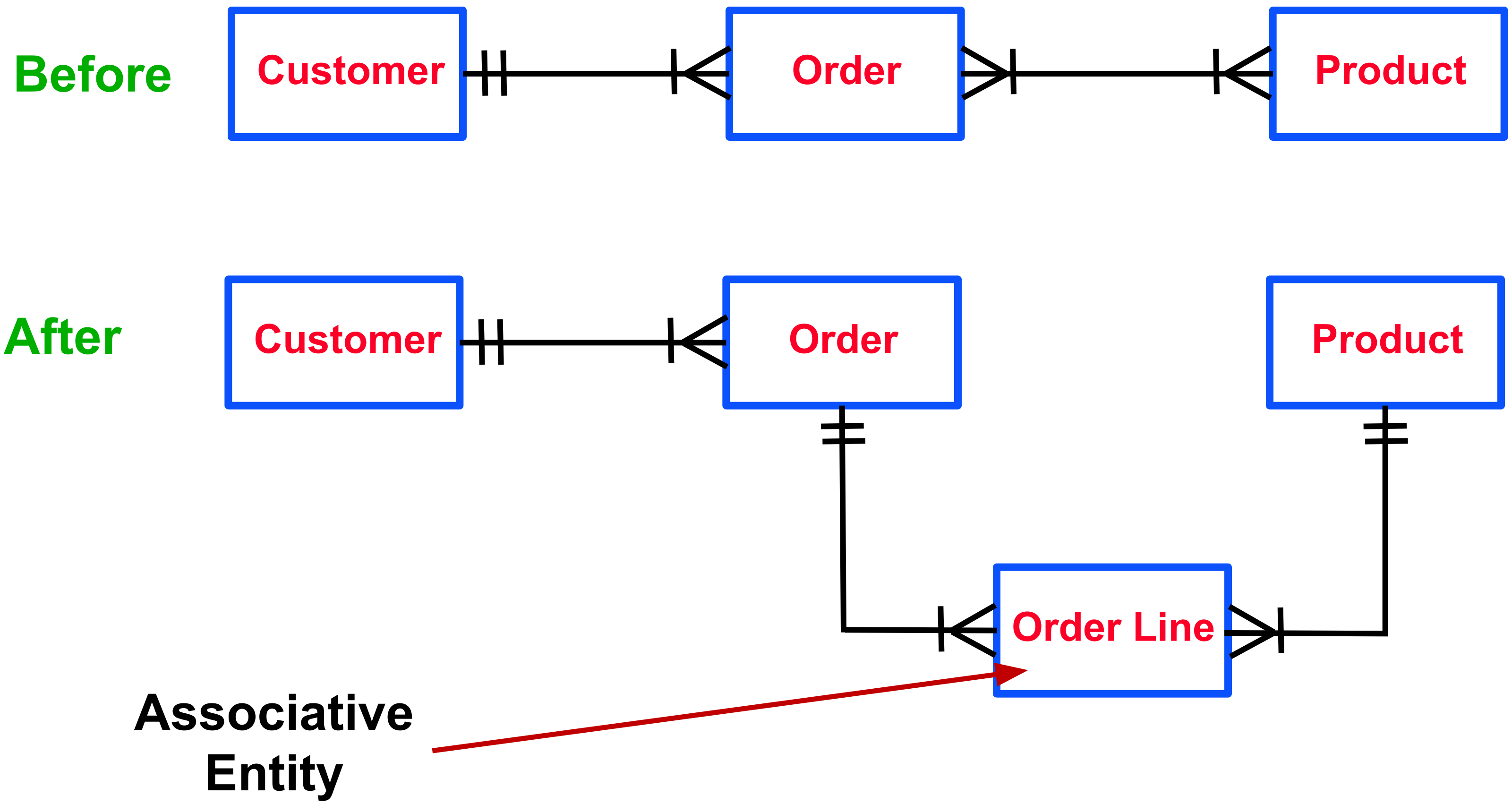
Crow's Foot Notation Examples

One SUPPLIER can supply one or many PRODUCT.
One PRODUCT can be supplied by only one

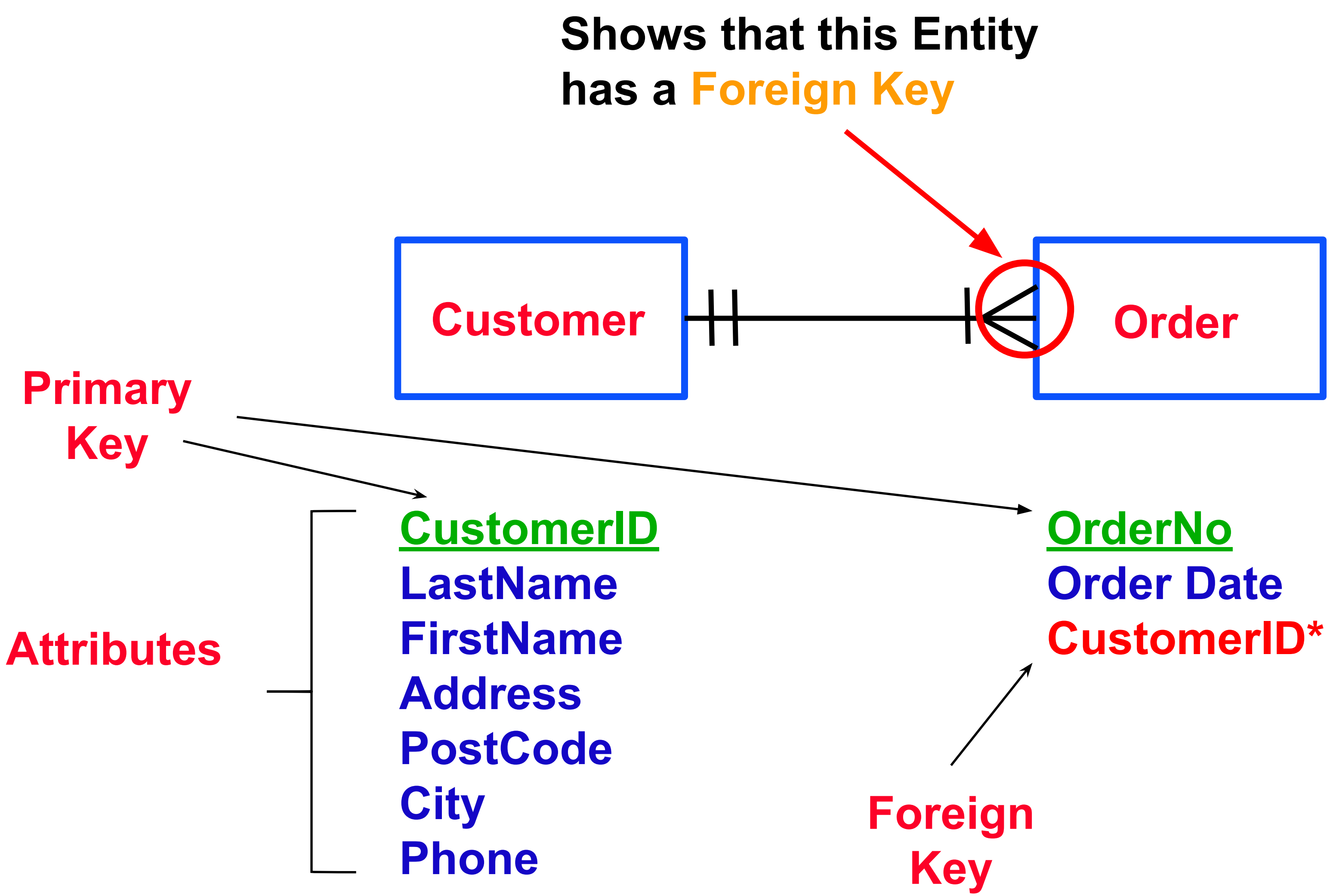
One MEMBER can borrow zero or one or many BOOK
One BOOK can be borrowed by zero or one or many MEMBER

Example 1 – Associative Entity

- By introducing an 'associative' entity to form a new **2 pairs** of one-to-many relationships.



Listing Data Attributes (Fields)



Sample Data for Each Attribute

Table Name

Attribute Names

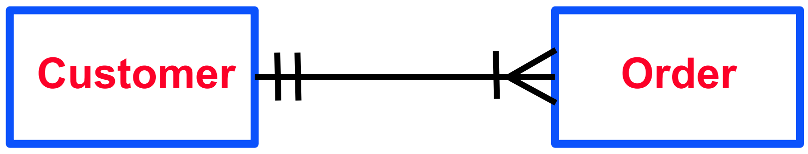
STUDENT

STUDENT-NUMBER	STUDENT-NAME	TOTAL-CREDITS	GPA
1035	Linda	47	3.647
3397	Sam	29	3.000
4070	Kelly	14	2.214

Data records

Listing Data Attributes in DBDL format

- DBDL = Database Design Language



CUSTOMER (<u>CustomerID</u>, LastName, FirstName, Address, PostCode, City, Phone)

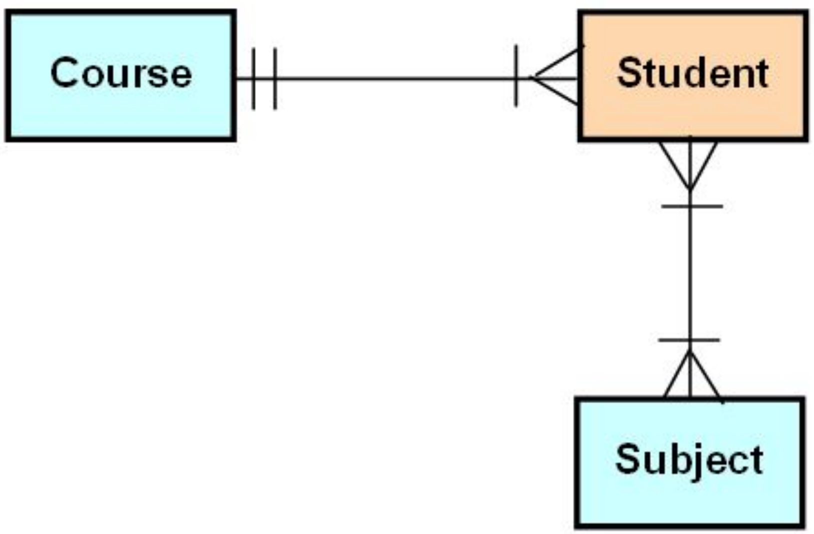
ORDER (<u>OrderNo</u>, Order Date, CustomerID*)

Primary Key

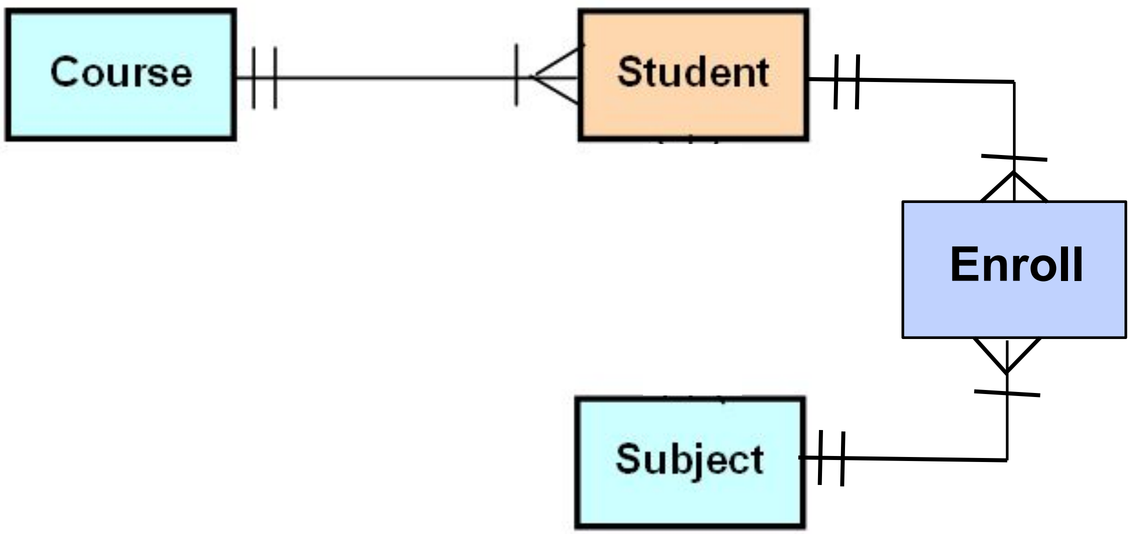
Foreign Key

Example 2

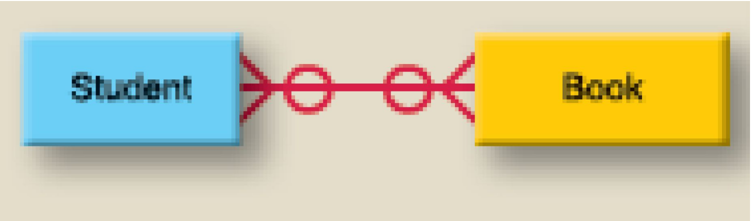
Before

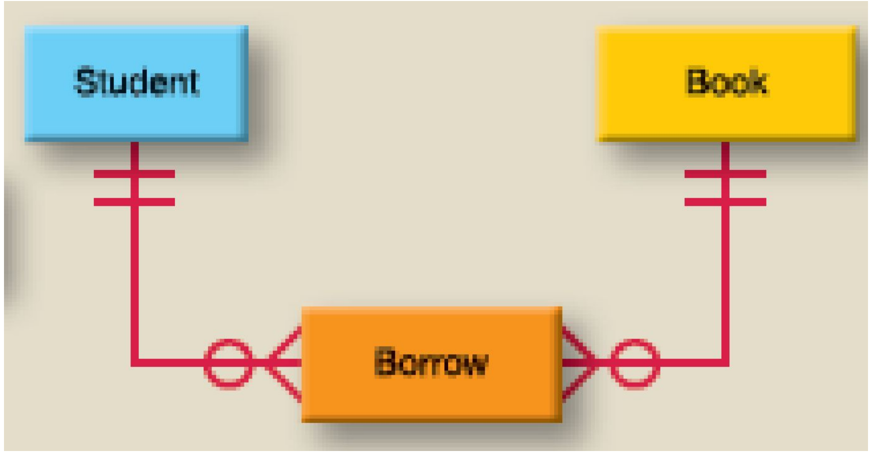


After



Example 3





Purposes of ERD

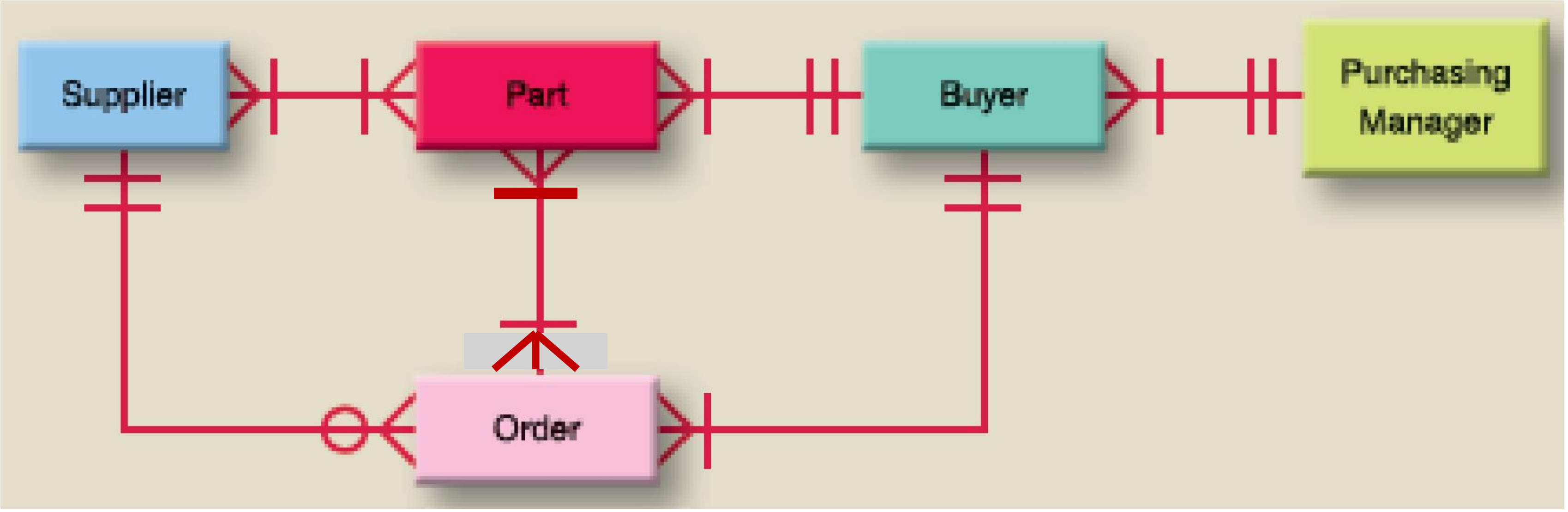
- **Business Rules** – capture and describe the business rules and relationships between the entities.
- **Modeling System** – describe the business system in diagrams.
- **Users** – communicating system to the users for their confirmation (sign-off) of requirements.
- **File Design** – form the basis of subsequent data and file design. The entities become physical files or database tables.

Exercise 1 - Purchasing System

- A company has 10 buyers, each one reporting to one of two purchasing managers.
- Each buyer has sole responsibility for the ordering of 500 of the 5000 parts used by the company.
- The same part can be supplied by several suppliers.
- Each purchase order can contain several parts.

Note : The above has been intentionally described in ONE direction only. Students should make logical assumption on the other direction.

Suggested Answer 1



Exercise 2 – Invoicing System

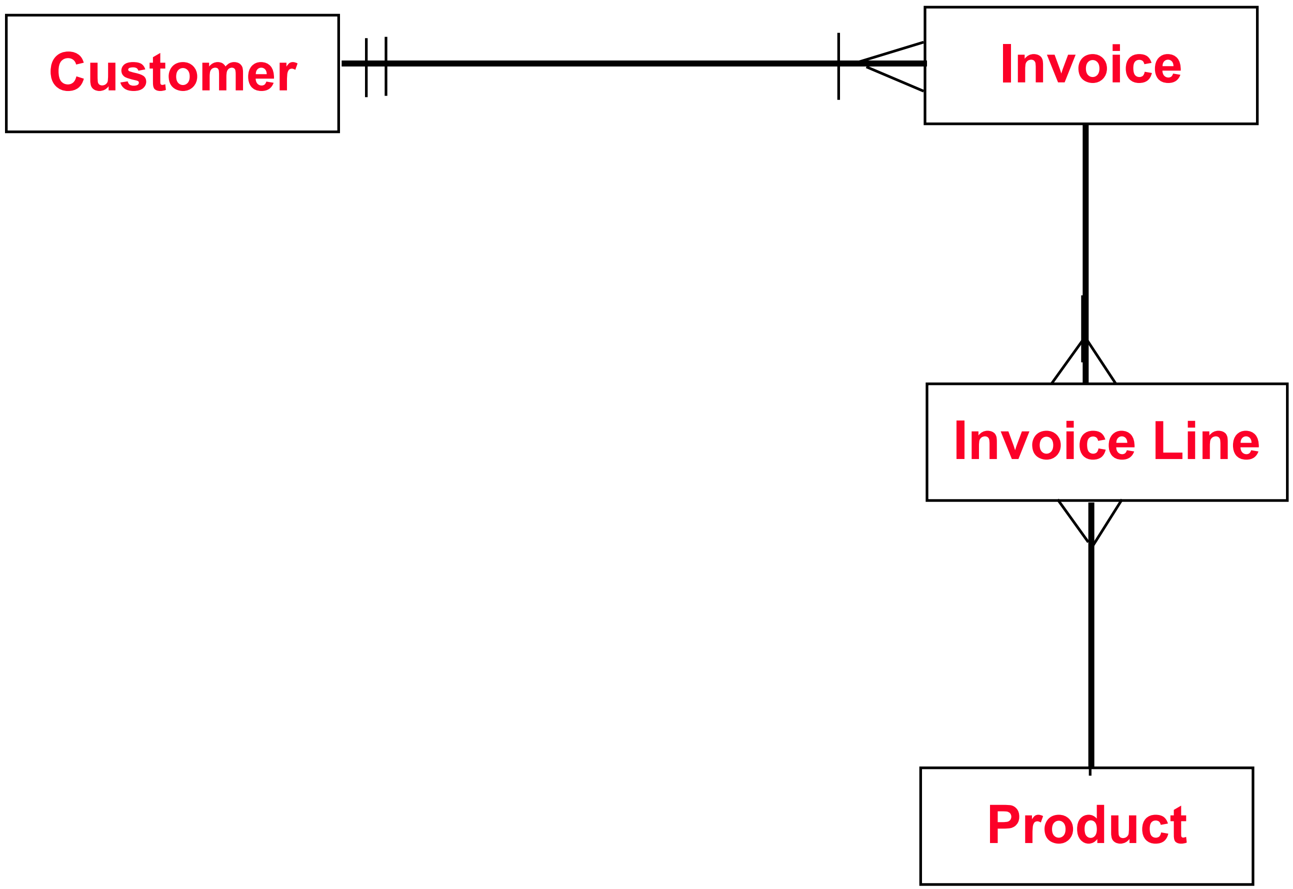
- Each Customer may generate one or more Invoices
- Each Invoice is generated by only one Customer.

- Each Invoice contains one or more Invoice Lines
- Each Invoice Line is contained in only one Invoice

- Each Invoice Line references only one Product
- Each Product may be referenced in one or more Invoice Lines

Note : Each relationship can be read in BOTH directions

Suggested Answer 2

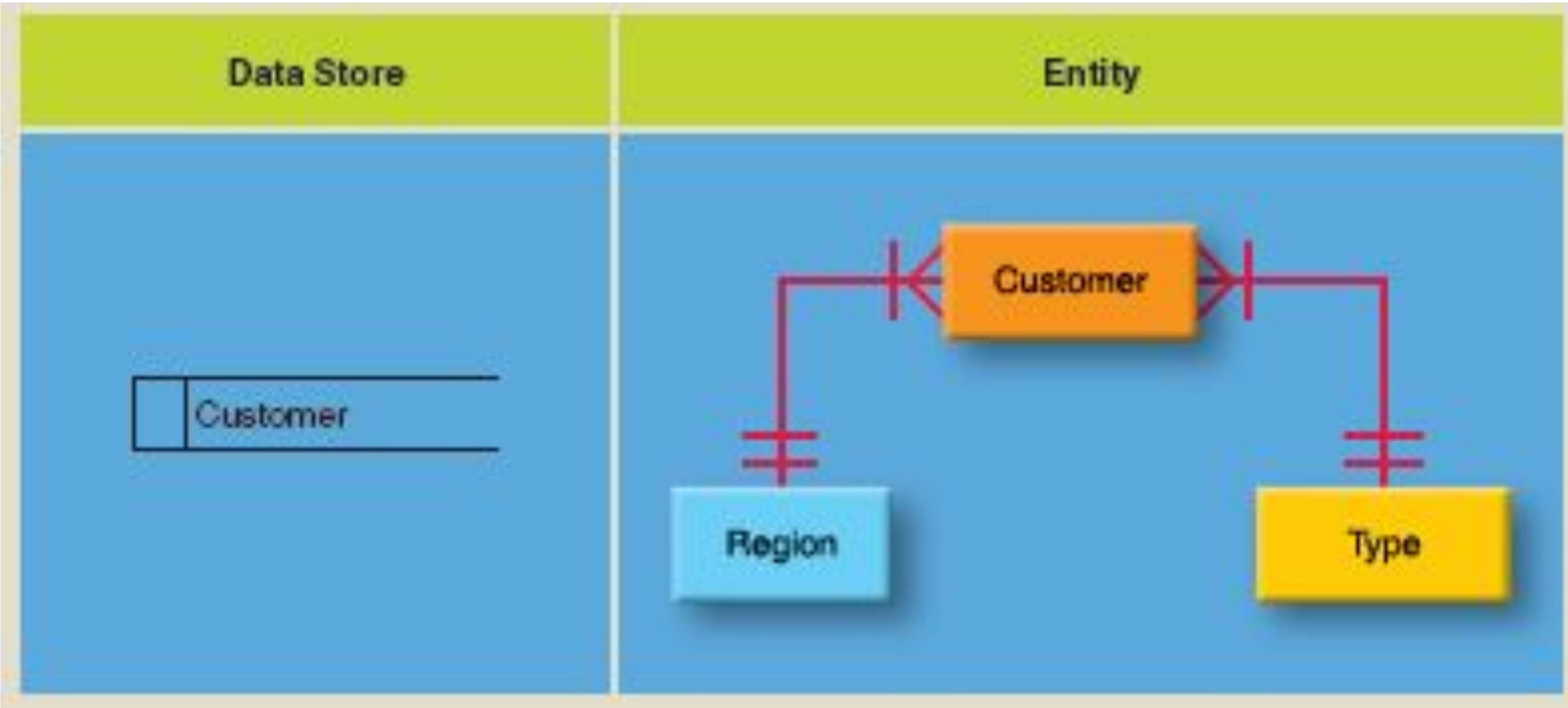


Describe the above relationships

Relationship Between DFD and ERD

- There is a relationship between DFD and the ERD. The data stores in DFD is related to entities in ERD.
- A well-drawn DFD can be used as a starting point for drawing the ERD.
- But take note that this relationship may not be 1 : 1. For example, the diary data store in DFD of a dentist system corresponds to the entities : appointment, appointment time and appointment type, in an ERD.

Relationship Between DFD and ERD
(cont…)

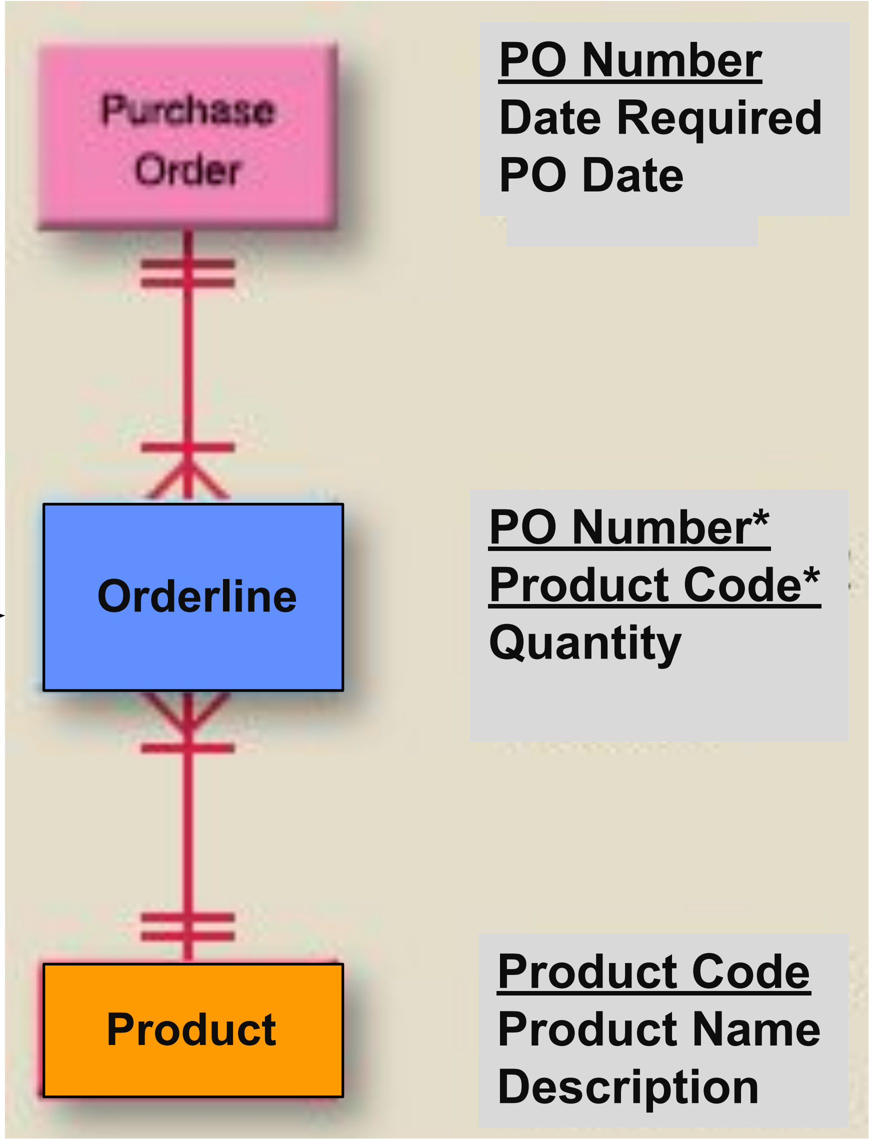


The **data stores** in DFD is related to **entities** in ERD. The number of data stores can be used as a starting point for drawing the ERD. But take note that this relationship may not be 1 : 1.

Exercise 3

- Indicate both the primary and foreign keys

Associative Entity



Exercise 4 for Students

- At the Best University, students enroll on courses (e.g. BA in Business Administration), but they also need to enroll on units that relate to these courses.
- A minimum number of students need to enroll on each unit for it to run.
- Each course has typically a number of units that can make up its total number of credit points.
- Each unit is led by one lecturer, but can be co-taught by other lecturers. Some lecturers get research-related leaves and so they don't need to teach on or lead any units.
- Also, each lecturer may have a number of students who are his or her "academic advisor" who can consult him or her on personal and other types of problems.

Answers 4